



Assessing the long-term ecological effects of riparian management practices on headwater streams in a coastal temperate rainforest



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ABSTRACT

Our understanding of how effective forestry Best Management Practices (BMPs) can mitigate logging impacts on the structure and functioning of temperate headwater streams tends to be based upon studies conducted within the first few years post-harvest. Little is known about how the affected biological parameters of streams change during riparian and upland forest regeneration, particularly the time taken to return to pre-harvest levels, or to within the range of reference conditions. By following up on an earlier study in southwestern British Columbia, Canada, we examined the post-logging recovery trajectories of stream attributes under multiple forest treatments. These treatments provided a gradient of riparian protection, including streamside clear-cutting with no reserves, and with fixed-width reserves (10 m and 30 m), and thinning with 50% basal area removal of riparian trees, which were compared with unharvested control sites. About 15 years after clear-cutting and 9 years after thinning, slower decomposition of red alder leaves (*Alnus rubra*) was reported in streams affected by forest harvesting, which was attributed to lower density of shredder invertebrates. Thinned reaches had lower litter decomposition rate than, but similar shredder density and richness to, unharvested reference sites. After a further 7 years of forest regrowth (time elapsed between the previous and present study), we found that litter decomposition rate and shredder density in logging-affected reaches converged with values in reference reaches. Stream functional integrity, as reflected by litter decomposition, increased across all forest treatments. Some differences in rarefied shredder richness among forest treatments emerged in this study. The legacy effects of thinning on some biotic variables likely diminished more quickly (2–9 years post-harvest) than those of clear-cutting with and without riparian reserves (8–15 years). Stream recovery from thinning to reference conditions should continue to warrant attention, as wider, larger-scale partial harvesting practices are recommended within emerging paradigms of forest management. Overall, our findings indicate time scales towards recovery of an important ecological process of streams under common forestry practices, which could be longer than other stream properties, such as water quality. Such differences in recovery time frames should be considered for the planning of forestry operations and monitoring of harvesting legacies.

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1. Introduction

A range of management practices have been designed to address and mitigate the impacts of forest harvesting on watersheds, which operate on various spatial scales. These practices typically include the establishment of riparian reserve strips of fixed or variable widths and configurations (reach scale); and harvesting of various extent and species (e.g., partial harvesting, progressive harvesting, selective and shelterwood logging), re-routing of

stream crossings and road networks, etc. (catchment scale). Many of these practices are now formulated as forestry best management practices (BMPs), and incorporated in forestry guidelines and regulations worldwide (e.g., Lee et al., 2004; McDermott et al., 2010; Cristan et al., 2016), as well as forest certification programs (Ice et al., 2010).

Contemporary forestry BMPs and other riparian protection measures are known to differ in the effectiveness of maintaining water quality, aquatic and riparian biodiversity, ecosystem functions and services (Cristan et al., 2016). Such effectiveness tends to increase with the width of riparian reserves, and fixed-width reserves of 30 m have been typically set to balance stream-riparian protection and timber production. Nevertheless, forest

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harvesting even ≥ 30 m beyond streambanks is sometimes shown to cause apparent abiotic (e.g., temperature and nitrate removal efficiency; see Sweeney and Newbold, 2014) and biotic changes in stream and riparian systems (e.g., Pearson and Manuwal, 2001; Kreutzweiser et al., 2008a; Lecerf and Richardson, 2010; Marczak et al., 2010). Partial harvesting of riparian vegetation (or thinning; usually affecting $\leq 50\%$ of the basal area) could elevate stream temperature (Macdonald et al., 2003; Guenther et al., 2014) and nutrient levels shortly after logging (Wang et al., 2006). The effects of thinning upland and riparian vegetation, or thinning upland forests while retaining riparian buffers, on stream habitats and communities may be evident (e.g., Martel et al., 2007; Moseley et al., 2008; Kara et al., 2014; Burton et al., 2016) or undetectable (e.g., Wilkerson et al., 2006; Olson and Rugger, 2007; Kreutzweiser et al., 2009; Chizinski et al., 2010), depending on the extent and methods of harvesting.

The effectiveness of riparian protection measures, such as fixed-width buffers and partial harvesting, has typically been evaluated based on short-term, post-harvest water quality data (e.g., within three years after harvesting; see Cristan et al., 2016). Post-harvest ecological recovery of streams, including biotic communities and their functions, has been much less studied, which can require longer time frames (but see Jackson et al., 2007; Gravelle et al., 2009). Many studies of ecological recovery adopted various sampling designs to minimize site-specific differences in ecological processes post-harvest, by studying streams affected by a gradient of logging history (Herlihy et al., 2005; Martel et al., 2007), multiple paired watersheds and three or more sites in treatment/control groups in replicated designs (e.g., Frady et al., 2007; Hemstad et al., 2008; Medhurst et al., 2010). Some studies detected signs of recovery for macroinvertebrate or shredder communities >10 years after harvesting (Davies et al., 2005; Frady et al., 2007; Medhurst et al., 2010), but did not address long-term changes in ecosystem processes, such as litter decomposition.

Measures of stream ecosystem processes are important to assess ecological integrity (Gessner and Chauvet, 2002). As forested headwater streams receive most of their energy inputs from allochthonous detritus, decomposition rates of leaf litter have been used to indicate forestry impacts on important ecosystem processes, with post-harvest increases (Griffith and Perry, 1991; Benfield et al., 2001; McKie and Malmqvist, 2009) and decreases (Kreutzweiser et al., 2008a; Lecerf and Richardson, 2010) being documented. The substantial variability of the post-harvest responses of litter decomposition rate (in terms of direction and magnitude) could be attributed to logging-induced alterations in stream habitats (Webster and Waide, 1982) and compositional shifts of riparian forest vegetation (Kominoski et al., 2011) (see also Richardson and Béraud, 2014). Several lines of evidence indicate that these changes affected the abundance and diversity of invertebrate shredders, as a key agent of organic matter breakdown in aquatic ecosystems (e.g., Griffith and Perry, 1991; McKie and Malmqvist, 2009; Lecerf and Richardson, 2010).

As part of the riparian management study at the University of British Columbia's Malcolm Knapp Research Forest (MKRF), near Vancouver, Canada, experimental harvests were executed across watersheds, with replicate streams for each harvesting practice (see Methods for detailed descriptions). The recovery of stream temperature and water chemistry in the logged sites to reference levels was detected within 5 years post-harvest (Gomi et al., 2006; Feller, 2010). In a study conducted in 2006, Lecerf and Richardson (2010) showed lower litter decomposition rates in streams 8 years after clear-cut logging with or without riparian reserves, and 2 years after thinning (i.e., logging 50% of the basal area of riparian trees) than in reference ones. The reduced decomposition rates in clear-cut reaches with or without riparian

reserves were associated with lower densities of shredder invertebrates.

We conducted this study in 2013 (i.e., 15 years after logging and 9 years after thinning) by re-measuring litter decomposition rate, shredder density and richness, and water quality characteristics in the same study sites. This study thus investigated the post-logging recovery trajectories of headwater streams in the presence of experimental riparian management practices. After an additional 7-year period of forest regeneration, we predicted that the measured biological parameters in some of the logged reaches, especially those with wider riparian reserves, would become close to, or no longer be significantly different from, those in reference reaches. Such convergence would indicate signs of ecological recovery from forest harvests, and would likely be associated with the altered physical environment of streams approximating reference conditions. We noted that several hydro-climatic characteristics, including temperature, precipitation and stream discharge, could potentially influence the responses of the measured biological parameters, in addition to the legacy effects of forest harvesting. For instance, stream discharge and thermal regimes in the study period and the preceding summer could affect shredder recruitment and survival, and hence litter decomposition, which need to be considered when comparing them across forest treatments.

2. Material and methods

2.1. Study sites

The set of study sites were the same as used in Lecerf and Richardson (2010). The study area is located within the Coast Mountains of the Pacific Northwest and about 45 km east of Vancouver, British Columbia. A total of 16 stream reaches (up to 5 m wide) in MKRF (49°18'40"N, 122°32'40"W) were chosen and randomly assigned (with one exception) to various riparian management practices, including (1) clear-cutting where no reserves were left; (2) upland clear-cut logging with 30-m continuous riparian reserves on both sides of the study reach; (3) upland logging with 10-m riparian reserves on both sides; (4) upland and riparian thinning (50% basal area removal). There were 3 or 4 replicate stream reaches in each riparian management practice (for details, see Kiffney et al., 2003 and Lecerf and Richardson, 2010). Experimental streamside logging with manipulations of fixed-width riparian reserves (0, 10-m and 30-m wide) took place in 1998, whereas thinning occurred in 2004. Thinning involved uniform tree removals spanning the stream channel within a harvest unit, with the goal of removing a similar basal area of trees on a catchment scale as in earlier upland logging.

Prior to the experimental harvests, these reaches and the unharvested, reference reaches were located in second-growth forests which had not been logged since 1931. Early successional riparian forests in the clear-cut sites was primarily dominated by rapidly-growing, nitrogen-fixing red alder, *Alnus rubra* Bong. (Betulaceae) in 2006 (Lecerf and Richardson, 2010). At the time of this study, detrital inputs of red alder (and other deciduous broadleaves) likely continued to dominate riparian litterfall to streams (Bilby and Bisson, 1992; Hoover et al., 2011), and enhance the productivity of lower trophic levels in clear-cut sites (Kominoski et al., 2012). Some red alder trees were also present in the riparian vegetation in sites with riparian reserves and thinned sites, and much fewer in coniferous-dominated reference sites. Canopy openness in each site was determined from digital hemispherical pictures taken vertically upward about 1 m from the streambed in autumn 2013 (Nikon Coolpix 4500 with fisheye lens, Nikon USA Inc., Melville, NY, USA). Pictures were processed using the software package

Gap Light Analyzer v 2.0 (Simon Fraser University, Burnaby, BC, Canada) to determine percent gap area within a 0–30° zenith range (Lecerf et al., 2016).

2.2. Water temperature and chemistry

Water temperature was recorded every 15 min throughout the deployment period of leaf bags in streams using temperature loggers (HOBO TidbiT v2 water temperature data loggers, Onset Computer Application). Temperature data were averaged for each day, in order to generate average daily water temperature for each stream during the deployment period. Measurements of pH and conductivity were conducted in situ at bag introduction and retrieval dates in all study sites. Water-quality parameters including total dissolved nitrogen (TDN) and soluble reactive phosphorous (SRP) were analyzed in the laboratory at the Great Lakes Forestry Centre at Sault Ste. Marie, Ontario, Canada, following standard methods (Beall et al., 2001).

2.3. Leaf bags

Senescent leaves of red alder were collected in autumn 2013, and then air-dried for several weeks. Nine leaf bags (made of 10-mm mesh size plastic netting), each containing 5 g of alder leaf litter (± 0.025 g), were deployed for each study site. Three sets of 3 bags were tied to iron bars, which were anchored into the stream bed in riffles of each of the study sites in early November 2013. A total of 5 additional leaf bags were submerged, and then immediately retrieved and transported back to the laboratory. The handling losses of leaf materials were estimated (mean: 7%; SE: 0.4%).

One set of 3 leaf bags were retrieved at each of 3 sampling occasions (mid-November, mid-December 2013, and mid-January 2014) from each stream. In the field they were stored on ice and returned to the laboratory for determination of remaining leaf mass. Retrieved litter was carefully removed from bags, and rinsed over sieves (500- μ m mesh size) to remove invertebrates and sediments from leaf surfaces. Leaf materials were oven-dried at 60 °C for 72 h and weighed to the nearest 0.01 g to obtain final leaf dry mass. Leaf tissue was then ashed at 550 °C for 4 h in a muffle furnace to determine ash mass, and hence ash-free dry mass (AFDM), for subsequent calculations of litter decomposition rates.

2.4. Shredders

Benthic macroinvertebrate shredders were sampled in early November 2013 prior to the introduction of leaf bags in streams. Five 900-cm² Surber (mesh size: 500 μ m) samples were taken from riffle habitats, immediately stored in 70% ethanol within plastic containers and transported to the laboratory. The mesh size of the Surber sampler was larger than the one used in the previous study (i.e., 250 μ m; Lecerf and Richardson, 2010), and a small fraction of smaller shredders thus might not have been captured in the sampling process. Benthic samples from each site were pooled together as in Lecerf and Richardson (2010). Shredders were counted and identified to the lowest practical taxonomic resolution to determine density and taxonomic richness.

Shredder richness needs to undergo individual-based rarefaction to properly correct for the number of shredders collected (see Gotelli and Colwell, 2001). Since the total number of shredders varied between sites and scaled positively with the number of taxa in both studies, taxonomic richness was uniformly rarefied to 11 individuals at each site. Sites with total abundance lower than 11 were excluded from the rarefaction (2 sites in Lecerf and Richardson (2010) and 3 sites in this study). Rarefied richness determined was more than 50% of total, unrarefied richness for all sites in both studies, hence inter-site differences in rarefied

richness were unlikely to be underestimated owing to the convergence of rarefaction curves that might occur at low abundances (Gotelli and Colwell, 2001).

2.5. Statistical analyses

Final leaf AFDM was adjusted by subtracting by the weight of leaf materials from handling losses. Daily and temperature-corrected litter decomposition rates were determined by obtaining the slope of the regression of the proportion of remaining leaf AFDM against time (in days) and degree days (sum of positive daily mean temperature), respectively. The ratio of decomposition rate coefficients (daily, non-temperature-corrected k values) in streams affected by each riparian management practice versus reference ones was calculated. This measure could reflect changes in stream functional integrity within the assessment framework proposed by Gessner and Chauvet (2002).

Differences in water characteristics (temperature, pH, and conductivity) across forest treatments were assessed by one-way multivariate analysis of variance (MANOVA), and those for biological parameters (litter decomposition rate, shredder density, and rarefied shredder richness) by separate ANOVAs. Two-way ANOVA was performed to detect differences in canopy openness across studies (i.e., years) and forest treatments. Planned comparisons further tested the hypothesis that logging impacts on biological parameters continued to be inversely related to riparian reserve width. These comparisons, using Helmert contrasts [reference vs mean (30-m reserves + 10-m reserves + no reserve), 30-m reserves vs mean (10-m reserves + no reserve), and 10-m reserves vs no reserve], identified groups which did not differ significantly ($p > 0.05$). These groups were then tested against thinned sites in the second set of contrasts.

Community dissimilarity of shredders between forest treatments for both studies was visualized using non-metric multidimensional scaling (NMDS) graphs, which showed centroids and 95% confidence ellipses associated with forest treatments. Shredders identified to be in the early instars stage (except the family Capniidae) in Lecerf and Richardson (2010) were excluded from the ordination analyses and richness calculations. Note these excluded taxa were still included in the calculations of shredder density. One-way permutational multivariate analysis of variance (PERMANOVA) was performed to assess whether the effects of forest treatments on shredder assemblages were different, using a Bray-Curtis similarity matrix. PERMANOVA is preferred as it is more robust to heterogeneity of variances (i.e., greater dispersion among groups) and unbalanced design than other widely used resemblance-based permutation methods such as analysis of similarity (ANOSIM; Anderson and Walsh, 2013), which apply to both studies. Complementarily, indicator species analysis (IndVal) was conducted to determine, for each taxon, whether it was a significant indicator of a particular forest treatment for both studies. The 'IndVal.g' value given is a statistic that measures the degree of association between a taxon and a forest treatment, which ranges from 0 (no association) to 1 (complete association; Cáceres and Legendre, 2009). $\ln(x + 1)$ transformations down-weighted the effects of abundant taxa on PERMANOVA and IndVal. The variability in assemblage structure (i.e., dispersion, or beta diversity) within each treatment group was measured by the average distance of site assemblages to the group centroid in multivariate space. Differences in dispersion among forest treatments for both studies were tested through the permutational analysis of multivariate dispersions (PERMDISP; Anderson, 2006) using the ANOVA F statistic. Post-hoc pairwise comparisons of group mean dispersions were made using a permutation-based test. All MANOVA and ANOVAs were conducted in R (R Development Core Team, 2015). All ordination analyses, including the construction

of 95% confidence ellipses (ordiellipses function), PERMANOVA (adonis) and PERMDISP (betadisper), were carried out using the vegan package in R (Oksanen et al., 2015). Indicator species analysis was performed using the 'multipatt' function in the indicpecies package in R (Cáceres and Legendre, 2009).

3. Results

3.1. Canopy openness and water characteristics

Percent canopy openness varied significantly among forest treatments (ANOVA: $F_{4,21} = 43.1$; $p < 0.001$) in this study. Canopy openness was generally higher in reaches without riparian reserves than with 10-m and 30-m reserves and thinned reaches, and it was the lowest in reference reaches. After a 7-year period of forest regrowth subsequent to the study by Lecerf and Richardson (2010), canopy openness declined overall (ANOVA: $F_{1,11} = 57.6$; $p < 0.001$) and was below 20% for all sites. On average, in clear-cut sites it had been reduced by half to 17%, and in 10-m reserves, by one-third to 13% (Table 1). For other treatments, the extent of openness decline over the 7-year period was much less remarkable, which was thus associated with a significant interaction effect between sampling years (i.e., time since logging) and forest treatments (ANOVA: $F_{4,11} = 15.0$; $p < 0.001$).

Variability of water characteristics among sites within a forest treatment was similar to, or greater than, their differences among treatments. These parameters were not significantly different overall across forest treatments (MANOVA: $F_{12,33} = 1.9$; $p = 0.08$), except for pH when analyzed individually (ANOVA: $F_{4,11} = 3.5$; $p = 0.045$). Notably, pH in clear-cut sites (mean: 5.7) was lower than that in 30-m reserves (mean: 6.4; Tukey's HSD: $p = 0.04$); however, all other pairwise comparisons showed no significant differences.

3.2. Litter decomposition

Decomposition rates for alder leaves were less variable across study sites both on a daily basis (0.0034–0.011 day⁻¹) and temperature-corrected basis (0.0006–0.0018 degree day⁻¹; Table 1) than previously measured (0.0019–0.0118 day⁻¹; 0.0003–0.0032 degree day⁻¹). Decomposition rates among logged reaches were similar regardless of reserve width, and did not differ significantly from thinned and reference streams (Table 2). The reduc-

tion of differences across forest treatments appeared to be attributable to considerable increases in decomposition in logged and thinned reaches (daily: 76–173%; temperature-corrected: 61–189%), and decreases in reference ones (25%; 46%), relative to those in the previous study (Fig. 1A and B). Stream integrity – as indicated by decomposition rate coefficients (Gessner and Chauvet, 2002) – improved across all forest treatments, which was no longer severely impaired and became either mildly or not impaired at all (Table 3).

3.3. Shredders

In this study, mean benthic shredder density was about 36–90% lower (range: 40.7–200 individuals m⁻²), and with smaller variability across forest treatments than those in the previous investigation (Fig. 1C). Densities of shredders at reference and thinned reaches were no longer significantly higher than those at logged ones ($p = 0.44$; Table 2), although the mean value at thinned sites tended to be more than double that of the other treatments (200 vs 40.7–82.2 individuals m⁻²; Fig. 1C, see also Appendix Table A.2).

When reanalysing data on shredder richness from Lecerf and Richardson (2010), rarefied richness actually did not vary significantly across forest treatments (ANOVA: $F_{4,9} = 1.18$; $p = 0.38$), and was incongruent with the results of unrarefied richness presented. Thus, only rarefied richness was analyzed and compared among studies (see Figs. 1D and A.1). In this study, rarefied richness also tended to be lower than in the previous study, with mean rarefied richness ranging from 2.1 to 3.4 taxa across treatments (Fig. 1D). Reference reaches in 2013 had significantly higher richness on average than logged and thinned sites, as indicated by the contrast analysis ($p = 0.01$; Table 2). Among logged sites, 30-m reserves supported the lowest richness compared to sites with 10-m and no reserves ($p = 0.01$). As for rarefied richness, the absence of *Yoraperla* stoneflies in most 30-m reserves and thinned sites (Appendix Table A.1), but not reference ones, could partly explain the lower richness in these logging-impacted sites in 2013.

The NMDS ordination plots represented data on shredder assemblage structure well, given the fair stress values (stress = 0.06 and 0.12; Fig. 2A and B). Differences in forest treatments did not significantly affect assemblage structures in the study by Lecerf and Richardson (2010) (PERMANOVA $F_{4,11} = 1.04$, $R^2 = 0.28$, $p = 0.39$, 999 permutations) nor the present study (PERMANOVA $F_{4,11} = 1.14$, $R^2 = 0.29$, $p = 0.32$, 999 permutations). This

Table 1

Reach-scale characteristics for the sixteen stream sites in the study. Water temperature during the study period (early November 2013 to late January 2014) is given as daily mean values. Other water characteristics obtained in the middle of the study period are shown. Soluble reactive phosphorous (SRP) concentration was below detection limit (i.e., 10 µg P L⁻¹) in all sites, and is therefore not shown in the table.

Site name	Forest treatment	Canopy openness (% open to sky)		Water temperature (°C)	pH	Conductivity (µS cm ⁻¹)	TDN (µg N L ⁻¹)	<i>k</i> (day ⁻¹)	<i>k</i> (degree day ⁻¹)
		2006	2013						
East	Reference	8.0	7.7	5.3	6.4	17	165.0	0.0086	0.0016
Mike	Reference	8.3	11.0	6.1	5.9	14	649.0	0.0034	0.0006
Spring	Reference	9.0	8.7	5.0	6.5	16	129.0	0.0079	0.0016
D	30 m reserves	12.5	13.3	6.5	5.6	13	66.0	0.0056	0.0011
H	30 m reserves	14.2	9.8	5.1	5.7	11	201.0	0.0052	0.0014
South	30 m reserves	13.5	12.0	6.2	5.8	13	79.0	0.0062	0.0009
C	10 m reserves	20.3	11.0	5.2	5.7	11	63.0	0.0086	0.0016
F	10 m reserves	21.8	15.5	6.9	6.3	15	66.0	0.0065	0.0013
G	10 m reserves	20.3	13.5	4.8	5.7	10	120.0	0.0060	0.0013
A	No reserve	33.2	16.9	4.2	6.5	14	238.0	0.0048	0.0008
B	No reserve	28.4	14.0	4.9	6.4	9	224.0	0.0054	0.0009
E	No reserve	–	17.4	6.1	6.2	11	<50	0.0058	0.0008
I	No reserve	36.8	18.9	3.3	6.5	10	217.0	0.0071	0.0018
J	Thinning	18.5	13.2	5.7	5.8	11	299.0	0.0107	0.0015
K	Thinning	19.9	16.0	4.7	6.5	16	127.0	0.0105	0.0017
L	Thinning	14.2	12.8	4.4	5.9	10	73	0.0056	0.0010

Table 2

Summary of ANOVA and planned comparisons of the effects of forest treatments on biological parameters. Shredder densities were ln-transformed to normalize the data and to improve the homogeneity of variance. Note that three sites were omitted in the analyses of rarefied shredder richness. *P* values are shown in parentheses, and are in bold when significant at 0.05.

Variable	All treatments (<i>p</i>)	Reference vs (30 m + 10 m + no) (<i>p</i>)	30 m vs (10 m + no) (<i>p</i>)	10 m vs no (<i>p</i>)	Thin vs reference (<i>p</i>)	Thin vs (30 m + 10 m + no) (<i>p</i>)
<i>k</i> (day ⁻¹)	$F_{4,11} = 1.5$ (0.27)	$t_{12} = -0.5$ (0.66)	$t_{12} = 0.7$ (0.52)	$t_{12} = 1.1$ (0.32)	–	–
<i>k</i> (degree day ⁻¹)	$F_{4,11} = 0.4$ (0.78)	–	–	–	–	–
Ln(shredder density)	$F_{4,11} = 1.0$ (0.44)	$t_{12} = -0.8$ (0.44)	$t_{12} = -0.2$ (0.86)	$t_{12} = -0.4$ (0.70)	–	–
Rarefied shredder richness	$F_{4,8} = 7.0$ (0.01)	$t_9 = -3.6$ (0.01)	$t_9 = 3.5$ (0.01)	$t_9 = <0.1$ (0.99)	$t_{11} = 3.2$ (0.01)	$t_{11} = -0.1$ (0.96)

was also revealed by the broad overlap of the 95% confidence ellipses for the centroids of forest treatments (Fig. 2A and B). In the present study, more shredder assemblages in sites affected by logging, both with and without riparian reserves, fell within the 95% confidence ellipse for the centroids of assemblages in reference reaches (Fig. 2A and B). Therefore, shredder composition in some logging-impacted sites began to be more similar to that in reference ones. Indicator species analysis showed, for the shredder assemblage in 2006, that *Moselia* (IndVal.g = 0.91, $p < 0.01$) and *Tipula* (IndVal.g = 0.86, $p = 0.03$) were significantly associated with reference and thinned sites. *Yoraperla* was representative of both 30-m reserves, reference and thinned sites (IndVal.g = 0.92, $p = 0.01$). However, no indicator taxa were found for any forest treatment in 2013, suggesting that taxa present became more commonly found across forest treatments (see also Appendix Table A.1).

Dispersion of assemblage structures differed significantly among treatments when re-analyzing the data from Lecerf and Richardson (2010) (PERMDISP; ANOVA $F_{4,11} = 5.38$, $p = 0.01$). Post-hoc comparisons revealed greater dispersion among clear-cut reaches than reference and thinned ones ($p < 0.01$; also see Fig. 2A). However, in the present study, significant differences in dispersion among treatments did not exist anymore (ANOVA $F_{4,11} = 0.39$, $p = 0.81$), as assemblages within clear-cut sites became more homogeneous, and had comparable within-treatment variability with that in other treatments ($p > 0.52$; Fig. 2B). This appeared to be partly associated with the occurrence of some taxa in more clear-cut reaches, such as *Despaxia* and *Lepidostoma* (see Appendix Table A.1), which led to the increase in similarity in shredder composition among sites. The dispersion of assemblages within sites influenced by thinning and in reference conditions was consistently low relative to other treatments across studies.

4. Discussion

We revealed that important biological parameters in streams affected by different forest treatments, such as litter decomposition and density of macroinvertebrate shredders, began to converge with reference conditions during the additional 7-year period of forest regrowth since the last investigation (Lecerf and Richardson, 2010). Successional riparian vegetation in logging-impacted streams experienced increases in canopy cover, although it was still less dense than that in reference sites. The denser streamside canopy mainly consisted of regenerating red alder stands and *Rubus spectabilis* Pursh (Rosaceae), a common shrub, particularly in no-reserve sites (Kiffney and Richardson, 2010). Winter stream temperature and water chemistry remained largely similar across forest treatments, except for TDN differences between thinned sites and the rest noted in Lecerf and Richardson (2010). This indicates that the recovery of abiotic variables, or water quality parameters, to reference levels was probably occurring in the previous study (i.e., within 8 years after logging and 2 years after thinning; see also Feller, 2010), which could precede that of litter decomposition rate and shredder den-

sity. This is also consistent with study findings elsewhere describing faster water quality recovery than ecological responses (e.g., Danehy et al., 2007; Kobayashi et al., 2010; Ishikawa et al., 2016). Plant cover and the biological uptake of essential elements normally increases rapidly in early forest succession following logging, thus revegetation would limit the export of sediments and nutrients (e.g., nitrates), which cease to affect water quality usually within several years after harvests (Webster et al., 2015).

4.1. Changes in shredder density, rarefied richness and assemblage structure

We found that shredder densities and rarefied shredder richness were uniformly, and often substantially, lower across all streams in the present study, when compared with Lecerf and Richardson (2010). While the effects of forest treatments on shredder densities appeared to have diminished, differences in rarefied richness across treatments were detected in this study. The emerging differences in rarefied richness between forest treatments was more likely attributed to changes in shredder diversity than in the evenness of relative abundance distributions of shredders (the latter could also affect the shape of rarefaction curves; Gotelli and Graves, 1996). This is because the evenness of the shredder community (represented by the Smith and Wilson evenness index E_{var} ; Smith and Wilson, 1996) was largely similar among treatments (range for 30-m reserves: 0.23–0.92; thinned sites: 0.28–0.90; clear-cut sites: 0.32–0.59; reference sites: 0.40–0.75; 10-m reserves: 0.91–0.92). The apparent reduction in shredder densities and rarefied richness across the two studies might be due to inter-annual climatic variability that influenced hydrologic disturbances (e.g., droughts, floods) in the summer or earlier months (Negishi and Richardson, 2006; Richardson, 2008), and/or other catchment-level conditions which affected the recruitment, survival and distribution of shredder communities (Hemstad et al., 2008; Kreutzweiser et al., 2010). The larger mesh size used in the present study (500 μ m) might partly lead to fewer shredders being captured. However, the extent of density reduction was likely to be small (<10–30%), even assuming that the early instars of leucotrid and nemourid stoneflies were all excluded (Lecerf and Richardson, unpublished data).

The return time of shredder densities to reference conditions was indistinguishable between clear-cut logging with and without reserves given our sampling regime, which could range from 8 to 15 years after harvests. The initial proportion of watershed harvested, along with other landscape features, could have long-lasting effects on the relative differences in litterfall, and benthic organic matter quality and quantity between clear-cut and reference streams (Frady et al., 2007; Ely and Wallace, 2010; Kiffney and Richardson, 2010). Importantly, these differences could influence consumer-resource interactions, and result in the apparent variability of shredder recovery across forest types. For instance, the effects of riparian and upland clear-cutting on shredders have been shown to diminish faster (<5 years post-harvest) in the sub-boreal interior region of British Columbia, Canada (Fuchs et al.,

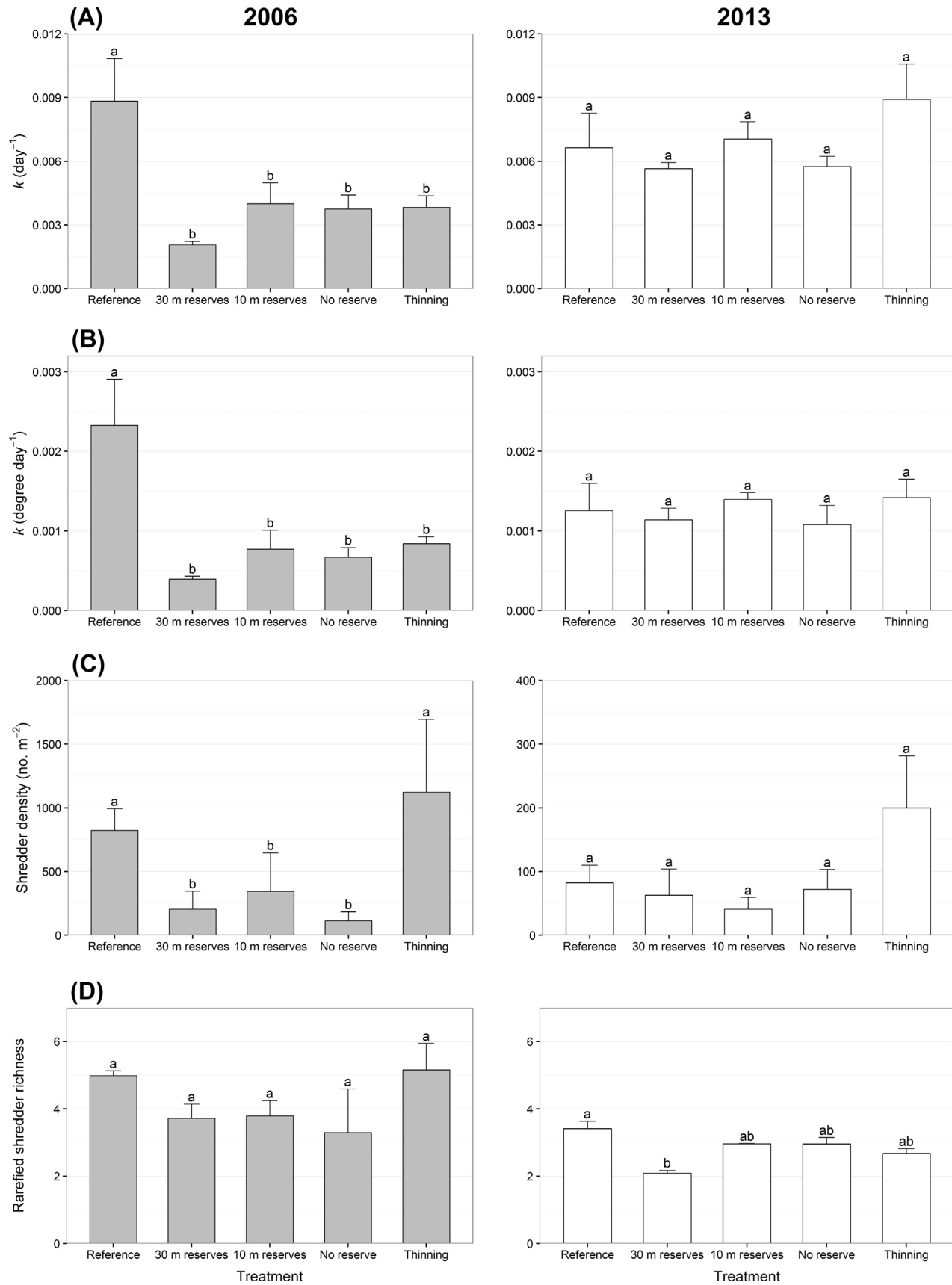


Fig. 1. Comparison of the effects of forest treatments on (A) daily and (B) temperature-corrected decomposition of alder leaves, on (C) shredder density and (D) rarefied taxonomic richness between studies conducted in 2006 (i.e., data from [Lecerf and Richardson \(2010\)](#)) and in 2013 (i.e., present study). Letters indicate homogeneous groups derived from ANOVA and planned contrasts uniquely for each study (see [Table 2](#)). Error bars denote standard errors. Note that scales of y-axis differ across studies in (C).

Table 3

Changes in the effects of forest treatments on functional stream integrity between studies conducted in 2006 (i.e., data from [Lecerf and Richardson \(2010\)](#)) and in 2013 (i.e., present study). The assessment of functional stream integrity was based on comparing daily, non-temperature-corrected decomposition rate coefficients at treatment (k_t) versus reference sites (k_r). Scores (in parentheses and bolded) represent the assessed conditions of the treatment sites (0: severely impaired; 1: mildly impaired; 2: no evidence of impairment). For details of assessment criteria, refer to [Gessner and Chauvet \(2002\)](#).

Forest treatment	k_t/k_r	
	2006	2013
No reserve	0.42 (0)	0.87 (2)
10-m reserves	0.45 (0)	1.06 (2)
30-m reserves	0.23 (0)	0.85 (2)
Thinning	0.43 (0)	1.34 (1)

2003), or a near-decadal time-frame in coastal western Oregon (generally <20 years; [Herlihy et al., 2005](#)) and also dry forests in central Washington, USA (mean: 16 years; [Medhurst et al., 2010](#)). The ranges of percentage of watershed area harvested in our study area (range for logged sites: 20–53%) were comparable with those in interior British Columbia (30–52%; [Fuchs et al., 2003](#)) and Oregon (0–42%; [Herlihy et al., 2005](#)).

When large portions or whole watersheds were clear-cut, the impacts of forest harvesting on shredders were typically shown to be longer-lasting in other studies. Shredder density in clear-cut streams was still substantially (~200–400%) higher than in reference sites 16 years after whole-watershed clear-cutting of the forest in the Coweeta Hydrologic Laboratory in North Carolina, USA ([Stone and Wallace, 1998](#)), but not 26 years post-harvest ([Ely and Wallace, 2010](#)). About 80% higher shredder density was recorded in streams draining young-growth forests even 40 years after clear-cutting than unharvested ones in the H.J. Andrews Experimental Forest in Oregon ([Fradley et al., 2007](#)). In temperate north-eastern Tasmania, Australia, taxon-specific responses of the abundance of facultative/obligate shredders 15 years after clear-cutting (50–90% of watershed area) were observed, with higher abundances of notonemourid stoneflies and leptophlebiid mayflies in clear-cut sites, but not for tipulid crane flies ([Davies et al., 2005](#)).

The legacy effects of riparian thinning on shredders or even macroinvertebrates appeared to be more short-lived than catchment-scale clear-cutting, as their numerical recovery in sites affected by thinning treatments (even in the presence of upland clear-cutting) could occur within 3–4 years post-harvest ([Kreutzweiser et al., 2010](#)), or sooner when only a single side of the stream was thinned ([Chizinski et al., 2010](#)). Thinning caused a relatively small reduction in canopy cover in these studies (<10%; see also [Kreutzweiser et al., 2009](#)), as was similarly found by [Lecerf and Richardson \(2010\)](#). Therefore, thinning appeared to cause lower overall disturbance to riparian forests and hence the shredder communities than that of clear-cutting.

Despite the statistical similarity of shredder assemblage structure across forest treatments, the assemblages in some logging-impacted streams were considerably different from those in reference sites in the previous study (see Fig. 3B in [Lecerf and Richardson, 2010](#)). After a further 7 years of forest regrowth, shredder taxa present could be found in most forest treatments. Also, assemblage dispersion among clear-cut sites, relative to that of reference ones, was reduced. This was associated with the re-appearance of certain shredder taxa (e.g., the stonefly *Despaxia augusta* and caddisfly *Lepidostoma* spp.) in clear-cut sites that were also present in reference ones. The mutual consideration of assemblage dispersion and overlap demonstrated subtle site-level variability in the delay of shredder responses to clear-cutting.

4.2. Changes in litter decomposition

In this study, we showed that some biological and physical parameters controlling decomposition rates in logged sites likely returned to within the range of variation of reference sites, although our sampling regime is not temporally intensive enough to confirm whether this had occurred earlier in reaches with wider riparian reserves. In thinned sites, faster decomposition was associated with lower TDN, relative to values in [Lecerf and Richardson \(2010\)](#), and both became comparable with reference conditions by 2013. The observed changes in decomposition rate did not appear to be related to any other water characteristics, because they continued to be similar among thinned and reference sites.

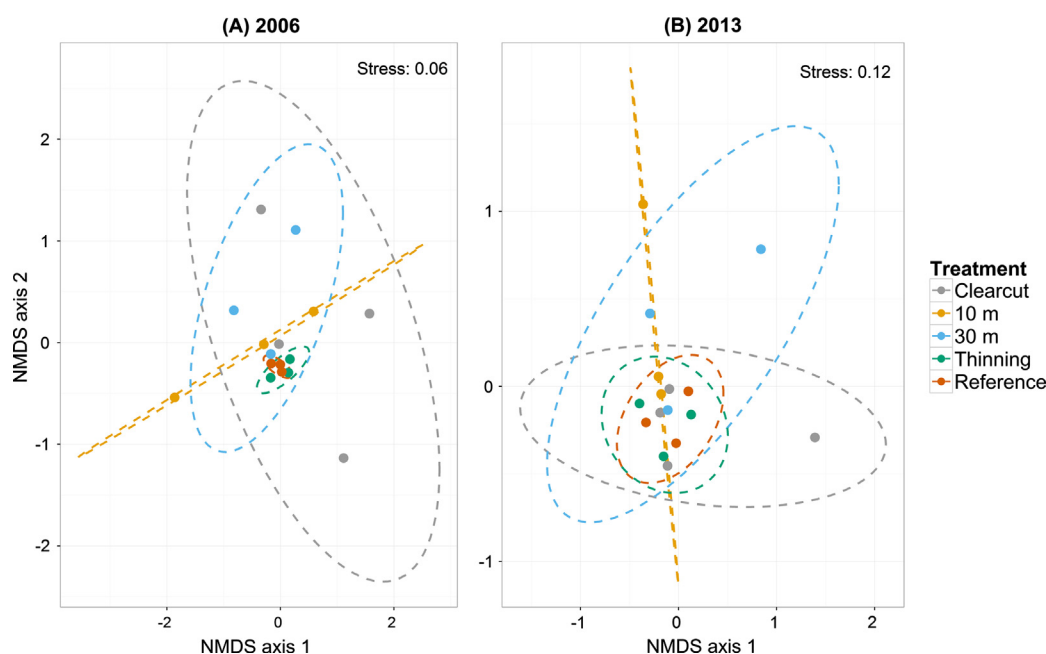


Fig. 2. Non-metric multidimensional scaling ordination of shredder assemblages across sixteen streams sampled in (A) 2006 and (B) 2013. Dashed ellipses represent 95% confidence intervals for the centroids of the groups of forest treatments.

The improvement of stream integrity across forest treatments observed in this study appeared to occur faster than in streams affected by catchment-scale clear-cutting in eastern United States, where decade-long post-harvest data are available. For instance, stream integrity was reported to remain moderately ($k_t/k_r = 1.42$; Griffith and Perry, 1991) to severely compromised ($k_t/k_r > 2$; Webster et al., 2014) even ≥ 20 years after harvests. The differences in recovery duration and trajectories of decomposition rates across forest types may arise from markedly variable responses of the underlying control processes of decomposition to riparian management histories. For instance, long-term increases in decomposition rate at clear-cut sites were associated with reduced benthic litter quantity and higher shredder density per unit area of litter (Griffith and Perry, 1991; Webster et al., 2014). Increased dominance of fast-processing leaves in riparian litterfall and/or reductions in total litterfall could lead to reduced availability of benthic organic matter in late fall and early winter. During the period of exacerbated resource limitation for shredders in detrital-based streams, litter in leaf packs could become an attractive food source and more rapidly consumed (e.g., Benfield et al., 2001; Rowe and Richardson, 2001; Tiegts et al., 2008). In our study, litterfall in some of the logged sites was approximating, or even greater than, that in reference ones within 8 years after harvests (Kiffney and Richardson, 2010), and we did not find higher shredder densities in logged and thinned reaches relative to reference ones, which unlikely intensified resource limitation. Elevated nutrient concentrations were identified as a cause of persistently higher decomposition rate in a stream 29 years after clear-cutting, within a montane deciduous forest in the Coweeta Hydrologic Laboratory (Webster et al., 2014); however, nutrient concentrations did not differ significantly among our study streams. Taken together, the return of some underlying drivers of litter decomposition to reference conditions may explain the absence of significant differences in decomposition rate across forest treatments.

4.3. Implications for forest management

The effectiveness of current forestry BMPs and other riparian protection measures has been predominantly evaluated on the basis of whether certain water quality standards are met in post-logging monitoring (see review by Cristan et al., 2016). The recovery of ecosystem structure (e.g., macroinvertebrate and fish communities) and functioning of streams may not necessarily be assumed even when there are no apparent signs of water quality impairment. Logging-induced changes in litter decomposition rates, and the quantity and quality of riparian litter inputs, could have long-term consequences on the local retention and downstream transport of particulate and dissolved organic matter (Kreutzweiser et al., 2008b). Importantly, these changes have the potential to affect the delivery of aquatic ecosystem services throughout the catchment, such as fish and habitat productivity (Bilby and Bisson, 1992; Wipfli et al., 2007). Therefore, we recommend the evaluation of harvest operations and forestry BMPs to look beyond water quality (see Karr, 1993), and consider the temporal scales of recovery for hydrologic, physiochemical and biological parameters from logging (see also reviews by Moore and Wondzell, 2005; Moore et al., 2005). Caution should be used when interpreting the recovery of shredder abundance and richness based on a few sampling times, in cases where the inter-annual variability of shredder recruitment and survival can mediate the legacy effects of logging. Commitment to long-term monitoring along the course of riparian and upland forest regeneration would be necessary to link forest regeneration to in-stream responses, and to verify the completeness of recovery to reference conditions, especially for catchment-scale clear-cutting.

Our study indicates that thinning of approximately 20% of the watershed area, including the riparian area, appeared to promote the rate of ecological recovery of streams, when compared to clear-cutting with and without fixed-width reserves in a coastal temperate rainforest in the Pacific Northwest. The sampling of two time points only characterized part of the post-logging recovery trajectory, and additional years of sampling are needed to confirm the recovery of the measured biological parameters from these forestry treatments. Thinning, or partial harvesting, near water bodies is often recommended in the emerging paradigms of emulating natural disturbance regimes (END; e.g., wildfires and windthrow) and enhancing future climatic resilience in forest management, by maintaining forest complexity and functioning (e.g., Sibley et al., 2012; O'Hara and Ramage, 2013; Hood et al., 2016). Indeed, this practice has been used on a commercial scale (usually affecting $\sim 50\%$ of the watershed area) to accelerate forest regeneration and sustain timber yields in the temperate and boreal regions (Curtis and Carey, 1996; Mielikäinen and Hynynen, 2003). Future efforts should therefore address how closely thinning at such scale could mimic natural disturbances in terms of the ecological legacies on headwater streams, as have been done for terrestrial biota (e.g., Schieck and Song, 2006; Zwolak, 2009; O'Donnell et al., 2015).

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.foreco.2016.10.044>.

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